A062 — Dynamic Vortex-Tube & Gravitational Source Closure

v0.5.0: 3D Periodic Euler/Navier–Stokes Evolution and Constitutive Gravity Gates

SST-Workbench | source-pinned technical falsifier

13 October 2026

1 Scope and provenance

The frozen instance maintains A062-v0.1.0–v0.4.0 as separate historical results and does not promote past physical failures. Two independently pinned E011 *static-ready* trefoil providers (Gilbert Fourier and KnotPlot) supply the initial centerline. Their finite-thickness Gaussian vortex tube is *not* a stationary Euler equilibrium. This v0.5.0 branch performs genuine nonlinear short-time three-dimensional *periodic* Euler and Navier–Stokes time integration. It does **not** establish long-lived knot dynamics or gravitational mass.

Automatic run provenance

Field	Recorded value
Run ID	20261013T182227899537Z-b96fbc73
Run phase	BLIND
Protocol bundle SHA-256	b4751339912f09b8b583ac379ac5c2951d4e369c2008cf31c2d4524686471358
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	a6e61d9d7e36e5007959e90328d3528cdd267bf5bbb3a3dd200e0791f2fff921

Gate ledger

Gate	Status	Question
G0	PASS	Did framework v1.0.8 freeze and validate protocol, provenance and local integrity?
G1	PASS	Are E011 Gilbert and KnotPlot independent original geometry SHA256 verified?
G2	PASS	Is Gaussian finite-width vortex tube energy convergent, positive and vorticity solenoidal?
G3	PASS	Does integrated kinetic energy yield a finite nonzero energy-equivalent mass candidate with
G4	FAIL	Does SST-73 organized stress double-divergence produce a free-space monopole for localize
G5	PASS	Does independent C++17 FP64 reproduce normalized finite Gaussian vortex-tube energy?
G6	NOT_APPLICABLE	Was optional SYCL/GPU kernel execution requested and hardware-certified? (Not applica
G10	PASS	Do independent 3D periodic short-time Euler/NS spectral simulations pass ABC, solenoida
G11	UNRESOLVED	Has long-term dynamically stable, topology-preserving, physically slender E011 trefoil been
G7	PASS	Do upstream-independent E011 geometries reproduce normalized tube energy within prede
G8	UNRESOLVED	Is physical free-space Swirl-Clock gravitational mediator, test-body universality and Newto
G9	DEFERRED	Was reveal separately authorized and executed?

Backend manifest


```
{
  "cpp_fp64": {
    "comparisons": [
      {
        "cpp_I": 1.6240955321895743,
        "provider": "gilbert",
        "python_I": 1.6240955321895743,
        "relative_error": 0.0
```

```
      },
      {
        "cpp_I": 1.8470697062769819,
        "provider": "knotplot",
        "python_I": 1.8470697062769816,
        "relative_error": 1.202145236698144e-16
      }
    ],
    "compiler": "C:\\workspace\\Strawberry\\c\\bin\\g++.EXE",
    "max_relative_error": 1.202145236698144e-16,
    "status": "PASS"
  },
  "limitations": [
    "Periodized large-width Gaussian tube != dynamically stable finite-core SST particle",
    "G10 PASS is a short-time solver screen only",
    "SST gravity, equivalence and G remain UNRESOLVED"
  ],
  "periodic_euler_ns_screen": {
    "metrics": {
      "abc_error": 2.207877364459415e-16,
      "max_NS_dissipation_budget_error": 1.7999240735176162e-06,
      "max_div_rel": 5.596644062076651e-16,
      "max_euler_energy_drift": 2.1295148795018792e-14,
      "max_initial_vorticity_projection_error": 5.746767893914135e-05,
      "max_spatial_E_diff": 3.954610013779985e-10,
      "max_temporal_E_diff": 2.161249798135477e-12
    },
    "pass_vector": {
      "abc_error": true,
      "max_NS_dissipation_budget_error": true,
      "max_div_rel": true,
      "max_euler_energy_drift": true,
      "max_initial_vorticity_projection_error": true,
      "max_spatial_E_diff": true,
      "max_temporal_E_diff": true
    },
    "status": "PASS",
    "thresholds_predeclared": {
      "abc_error": 1e-10,
      "max_NS_dissipation_budget_error": 0.005,
      "max_div_rel": 1e-10,
      "max_euler_energy_drift": 0.0005,
      "max_initial_vorticity_projection_error": 0.03,
      "max_spatial_E_diff": 0.02,
      "max_temporal_E_diff": 0.002
    }
  },
  "python_reference": {
    "actual_backend": "python-numpy-fp64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
  "scientific_boundary": "Actual projected 3D periodic Euler/NS time integration has only been tested in
  "sycl_gpu": {
    "authority": "SCREENING_ONLY",
    "requested": false
  }
}
```

Source manifest


```
{
  "manifest_sha256": "3afa391c1f73153e602078784c4fe8b8d5b981d7ef885adcf581e03aa7b92497",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "E011",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "gilbert",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A062_SST_Microgravity_
      "provenance_family": "gilbert_ideal_A004",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "discovery",
      "sha256": "274a4ce111607a6295e280f3f8369c665e7de82ac33afcd4cb227eb61568d530",
      "source_id": "S_GILBERT",
      "version": "0.3.0"
    },
    {
      "catalog_id": "E011",
      "evidence_class": "independent_source",
      "exists": true,
      "independence_family": "knotplot",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A062_SST_Microgravity_
      "provenance_family": "knotplot_relaxed_A001",
      "registry_hint": {
        "catalog_id": "E011",
        "hits": []
      },
      "role": "confirmation",
      "sha256": "8fea8a15ff3047dbbfb4aba05090b9a6cadc10eeedf97c22350925c764bc540e",
      "source_id": "S_KNOTPLOT",
      "version": "0.3.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


2 Equations and numerical method

For a bandlimited divergence-free velocity field $\mathbf{u}$ on a periodic three-torus $\mathbb{T}^3_L$ with $L = 1.2$ dimensionless, solve

$$\partial_t \mathbf{u} = \mathbb{P}(\mathbf{u} \times \boldsymbol{\omega}) + \nu \Delta \mathbf{u}, \tag{1}$$

$$\boldsymbol{\omega} = \nabla \times \mathbf{u}, \qquad \nabla \cdot \mathbf{u} = 0. \tag{2}$$

Here $\mathbb{P}$ denotes the Helmholtz–Leray Fourier projector. The nonlinear Lamb vector is pseudo-spectral with strict 2/3 de-aliasing; a fourth-order explicit Runge–Kutta integrator advances the field. No unverified external forcing is added.

Initialize each E011 source from the Gaussian-smoothed closed curve,

$$\boldsymbol{\omega}_a(\mathbf{x}) = \Gamma \oint_\gamma d\mathbf{X}\, (2\pi a^2)^{-3/2} \exp[-|\mathbf{x} - \mathbf{X}|^2/(2a^2)], \tag{3}$$

using $a = 0.11$ in a dimensionless box. Importantly, this **strong smoothing** exceeds the original upstream geometric reaches (roughly 0.025–0.028); thus the run is a *regularized dynamical screen*, not a fully resolved slender SST core. Projection/cutoff error is reported, not silently removed.

3 Analytic invariants and control solution

The discrete solver is measured against an exactly nonlinear-force-free Beltrami field $\nabla \times \mathbf{u} = \mathbf{u}$ with $L = 2\pi$. For unforced NS with $\nu > 0$,

$$\mathbf{u}(t) = \mathbf{u}(0) \exp(-\nu t). \tag{4}$$

For the periodic unforced system, in normalized units,

$$E = \frac{1}{2} \int_{\mathbb{T}^3} |\mathbf{u}|^2 dV, \quad \frac{dE}{dt} = -\nu \int_{\mathbb{T}^3} |\boldsymbol{\omega}|^2 dV. \tag{5}$$

At $\nu = 0$, both energy and helicity must be invariant for smooth solutions. Each E011 provider undergoes $N = 24, 32, 48$ resolution sweeps (FULL/CERTIFY) with $\nu \in \{0, 0.005\}$ and $T = 0.03$; $\Delta t = 0.002$ versus 0.001 supplies time refinement. The NS viscosity is *only* a screening parameter, not a fundamental SST viscosity.

4 Gravitational sources remain distinct

For localized stress $\Sigma_{ij} = \rho_f u_i u_j$, the SST-73 proposal has

$$S_C = \lambda \partial_i \partial_j \Sigma_{ij}, \quad \int_{\mathbb{R}^3} S_C dV = 0 \tag{6}$$

under adequate decay. On the periodic numerical domain, the same double divergence has exactly zero Fourier mean, $\widehat{S}_C(\mathbf{0}) = 0$. A periodic scalar Poisson equation $\Delta\Phi = K\rho_E/c^2$ is *inconsistent* for strictly positive mean source until background neutrality or a different boundary problem is specified. A nonzero hydrodynamic E/c^2 is **not** an independently derived universal gravitational source. No long-range $1/R$, equivalence principle, G normalization or microgravity test-body response has been obtained.

5 Falsification policy and unresolved gates

G10 assesses short-time numerical validity only: exact ABC error $< 10^{-10}$, div residual $< 10^{-10}$, Euler energy drift $< 5 \times 10^{-4}$, NS budget error < 0.005, time-refinement energy error < 0.002, last-grid energy error < 0.02, and raw-to-projected vorticity change < 0.03. Any failure triggers G10 FAIL but downstream gates still execute. G4 is an independent physical zero-monopole test. G6 is reserved by the framework for optional SYCL and remains NOT_APPLICABLE in the CPU profile. G11 requires long-term topology-preserving dynamic stability and remains UNRESOLVED. G8 requires a first-principles clock mediator, passive/source equivalence, free-space response and dimensionful G, and must stay UNRESOLVED unless derived. E011 geometry is STATIC_READY, not DYNAMICS_READY: topology-preserving transport, finite-core relative equilibria, reduced Floquet spectrum and nonlinear stability are outside the present screen.

6 Reproduction and integrity

The original SST Falsifier Framework v1.0.8 LOCAL_VALIDATED_CANDIDATE is included unchanged. FREEZE first creates a 32-byte instance-private nonce and opaque key *only* in a newly unfrozen instance with no prior commitments. Private secrets are never included in source releases. Framework output ZIP/PDF files use unique run IDs. The framework's known provisional-versus-final manifest hash issue is a report-ordering issue; individual final ZIP hashes must be verified separately.

7 Sources and epistemic scope

E011 Gilbert and KnotPlot independent raw sources; SST-18 (Euler vs mediator), SST-34 (short-range pressure interaction deprecated as long-range gravity), SST-59/61 (mass and topology), SST-73 (organized stress source) and SST Canon v0.8.35 (conditional energy/clock bridge, explicitly calibrated G). Indexed-only SST manuscripts, the excluded SST-28 model and announced Euler/NS singularity proofs are *not* used as empirical validation of this simulation.